Quality of Service (QoS) & Traffic Shaping

Coverage: QoS Metrics, Policing vs Shaping, Leaky Bucket vs Token Bucket, Scheduling (FIFO, WFQ), DiffServ vs IntServ

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1. Traffic Shaping

- Leaky Bucket: Rigid constant output rate rho.
- Token Bucket: Allows regulated bursts up to bucket capacity b at wire rate R for duration T = b / (R - r).

2. DiffServ vs IntServ

- DiffServ: Scalable packet classification using 6-bit DSCP markings at boundary routers.

■ Worked Problem: Token Bucket Burst Duration

- Given: b = 16 MB, r = 2 MB/s, R = 10 MB/s. T = 16 / (10 - 2) = 2.0 seconds. Total Data = 20 MB.

■ KTU University High-Yield Exam Questions

- Q: Compare Leaky Bucket and Token Bucket traffic shaping algorithms. (8 Marks)